

JEE MAIN REVISION MASTER: CARBOXYLIC ACIDS AND ITS DERIVATIVES 2026 1 QUESTIONS

1.67% WEIGHTAGE ↓49.85%

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 11 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Carboxylic Acids And Its Derivatives - Core Application	11 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Carboxylic Acids And Its Derivatives - Core Application (High Yield — 11 PYQs)

Core Principle: Key problem pattern in Kcet Carboxylic Acids And Its Derivatives.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Carboxylic acids are more acidic than phenols because..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "Statement - I : Reduction of ester by DIABL-H followed by hydrolysis gives aldehyde. Statement - II : Oxidation of benzyl alcohol with aqueo..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "Which of the following has most acidic Hydrogen ?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Carboxylic acids are more acidic than phenols because

- (A) Formation of dimers
- (B) Intermolecular hydrogen bonding
- (C) More covalent nature
- (D) More resonance stabilisation of their conjugate base

Correct Answer: None

Solution Analysis: Carboxylic acids are more acidic than phenols because their conjugate base is more resonance stabilised. So the correct option is: Option D: More resonance stabilisation of their conjugate base. When a carboxylic acid loses a proton, it forms a carboxylate ion: $\text{RCOOH} \rightarrow \text{RCOO}^- + \text{H}^+$. In the carboxylate ion, the negative charge is delocalised equally over two oxygen atoms: $\text{R}-\text{C}(=\text{O})-\text{O}^- \leftrightarrow \text{R}-\text{C}(\text{O}^-)=\text{O}$. Both resonance structures are equivalent, so the carboxylate ion is highly stable. Phenol also forms a conjugate base, called phenoxide ion: